

CAST: Testing the Attribution Stability of AI-Generated-Code Detectors Under Controlled Source Transformations

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Abstract— AI-generated-code detectors are evaluated almost exclusively on static snapshots, even though real code is routinely reformatted after creation, for example by an IDE on save. We present CAST (Code Attribution Stability Testing), a controlled methodology that measures whether a detector’s human-versus-AI attribution survives such changes. On 199 held-out Droid programs, reformatting with a standard tool (`black/google-java-format`) flips a reconstructed detector’s (DroidDetect-Base) correct verdict on genuinely human-written code to “AI” 58.0% of the time, versus 0.0–4.3% on actually AI-authored code; a reverse check, messifying AI code, induces only a 5.2% AI-to-human flip, though the two perturbations are not intensity-matched. This asymmetry is consistent with the detector having learned tidy formatting as an AI-authorship proxy, false-accusing human code, with direct stakes for students, job candidates, and contributors under AI-disclosure policies. LLMsSniffer’s low error rate is not robustness evidence: it is at or below a majority-class baseline on this corpus. Detect-CodeGPT’s score movement is within its own re-scoring noise floor. Canonicalization attenuates DroidDetect-Base’s residual movement by 14.8–15.4 $\times$, but canonicalized input is itself often out-of-distribution, so this diagnostic does not yet establish a mitigation. Static accuracy alone does not characterize detector reliability under routine code evolution.

Index Terms—AI-generated code detection, robustness testing, metamorphic-style evaluation, software evolution

I. INTRODUCTION

AI coding assistants now write a substantial share of the code developers review, edit, and ship. Detectors that attribute this code to a human or AI origin are typically evaluated the same way a classifier is: on a fixed test set, scored once, with accuracy as the verdict.

Motivation: That protocol quietly assumes the code being judged stays put. It does not. An IDE reformats a file on save, a refactor renames a variable, a reviewer rewrites a loop – none of it changes what the program does, yet any of it can change what a detector says about who wrote it. A detector whose verdict flips under a cosmetic edit is not measuring origin; it is measuring the snapshot – with real consequences when that verdict feeds an academic-integrity case, a hiring assessment, or a project’s AI-disclosure policy. This is a routine, non-adversarial threat distinct from the adversarial setting the closest prior work targets: robustness studies vary how code is generated [11] or train against an attacker’s

renaming [6], [1], and separate work shows renaming can defeat code watermarks [7] and that code-model metamorphic testing treats renaming as standard [8]. None measures whether an already-trained detector’s verdict survives a non-adversarial edit, or reports which edits matter and why.

Objective: This paper’s objective is to make attribution stability an explicit, measurable property of an AI-code detector, not an afterthought to accuracy, and to build the smallest instrument that can measure it. We introduce **CAST** (Code Attribution Stability Testing): given a program of known origin, CAST applies a validated transformation, measures the resulting change at the text, token, and AST level, and records how the detector’s score and decision move (Section II). Where movement is large, CAST adds a second, targeted step: canonicalizing both sides of the pair to test whether the instability is a presentation artifact or something deeper. Applied to three detectors over 199 held-out Python and Java programs [1], this approach exposes what a single accuracy number cannot: one detector shows a large, class-specific, false-accusing instability under formatting that canonicalization only partially and ambiguously attenuates; the other two detectors’ apparent profiles turn out to be confounded by degenerate baseline accuracy and re-scoring noise, a finding about evaluation practice in its own right.

This paper makes two contributions.

- **CAST**, a controlled, distance-instrumented diagnostic methodology for evaluating the attribution stability of already-trained AI-code detectors under routine, non-adversarial edits, including a canonicalization intervention and an identity-noise-floor control.
- A held-out evaluation showing one detector’s large, validated, false-accusation-dominated instability, and a methodological lesson that a low IER is uninterpretable without checking baseline accuracy and re-scoring noise first.

The remainder of the paper is organized as follows. Section II defines CAST and the canonicalization intervention. Section III describes the dataset, transformations, detectors, and research questions. Section IV reports results; Section V discusses their implications. Section VI lists threats to validity, Section VII positions CAST against related work, Section VIII outlines future plans, and Section IX concludes.

II. CAST

A. Attribution stability

Attribution stability is the degree to which a detector’s output for a program is preserved under routine, non-adversarial edits – distinct from adversarial robustness, which studies a worst-case attacker searching for an evasive rewrite. CAST’s transformations are fixed, deterministic operators applied uniformly across the corpus, not an adversarial search against any detector. CAST treats score movement and decision-flip rate as first-class observations, not a single post-transformation accuracy number.

B. Methodology

For a detector D , a transformation T , and a program C_0 of known provenance, CAST proceeds in five steps.

- 1) Apply T to obtain $T(C_0)$.
- 2) Validate that $T(C_0)$ is well-formed at or above C_0 ’s own syntactic-validity tier. Non-applicable and invalid outcomes are recorded, not discarded.
- 3) Measure d_{ext} (character-level edit distance), d_{token} (edit distance over the language’s own lexical token stream, e.g., Python’s `tokenize` output including `INDENT/DEDENT`, or Java tree-sitter leaves), and d_{AST} (AST node-count change).
- 4) Score $D(C_0)$ and $D(T(C_0))$.
- 5) Aggregate, per detector, language, and family, the mean and median score movement $\Delta_D = D(T(C_0)) - D(C_0)$, the decision-flip rate, and the **Induced Error Rate**, defined as:

$$\text{IER} = \frac{\#\{\text{baseline-correct programs whose decision flips after } T\}}{\#\{\text{baseline-correct programs}\}} \quad (1)$$

IER is conditioned on the detector having been correct on C_0 . A flip on a program the detector already misclassified would otherwise conflate ordinary detector error with transformation-induced instability. Detectors also expose incompatible score semantics: a calibrated probability, an uncalibrated sigmoid output, or an unnormalized statistic. CAST therefore does not compare raw $|\Delta_D|$ across detectors. IER, a bounded proportion with a consistent interpretation regardless of score scale, is the primary cross-detector metric. Raw score movement is interpreted only within a given detector’s own distribution.

C. Canonicalization intervention

When a transformation family produces large score movement, CAST additionally scores $F(C_0)$ and $F(T(C_0))$ for a deterministic canonicalizer F that removes a specific, named category of variation (in this study: whitespace, blank lines, comments, and quote style, implemented by re-tokenizing the program and rejoining its semantic tokens). A pair for which $F(C_0)=F(T(C_0))$ collapses to $\Delta=0$ by construction. This subset is reported separately from the residual pairs for which a genuine difference remains after canonicalization, so that the diagnostic result is not an artifact of the trivially-resolved cases. A substantial reduction in the residual subset’s score movement constitutes intervention evidence that the removed

category of variation contributes to the observed instability. It does not constitute proof of a specific internal mechanism.

III. STUDY DESIGN

Dataset: This evaluation uses DroidCollection’s [1] held-out test split: 199 programs (100 Python, 99 Java), stratified across HUMAN_GENERATED, MACHINE_GENERATED, and MACHINE_REFINED.

Transformations: Three confirmatory families: formatting (`black/google-java-format`), lexical renaming of one identifier, and control-flow rewriting (one canonical loop-form rewrite), with applicability rate 0.616 and success rate 0.992 among applicable attempts. A fourth stage attempted two deeper families, local structural extraction and semantics-preserving rewriting, reported as a feasibility finding (Section IV-C), not a confirmatory one.

Validation: Python uses `compile()`-based validation. Java is tiered, given the corpus’s many function/class fragments: `STANDALONE_VALID` > `WRAPPER_VALID` (compiles inside a synthetic harness, used only as a validation instrument and never scored) > `STRUCTURAL_ONLY` > `PARSE_INVALID`.

Detectors: Two of five targeted detectors had a released checkpoint runnable as-is; a third (DroidDetect-Base) required reconstructing its architecture from the checkpoint alone. *LLMSniffer* [2] (GraphCodeBERT [19], supervised-contrastive) reaches 90.0% on GPTSniffer’s Java test subset [3] but only 67.0%/44.4% on this corpus (at or below majority-class baseline; Section VI). *DroidDetect-Base* (ModernBERT-base [18] with a projection layer and a four-class head) ships a checkpoint with no modeling code; we reconstructed its architecture and validated it against a disjoint dev-split subset ($n=24$), reaching 87.5% accuracy (Wilson 95% CI [69.0,95.7]). This does not fully contain the 98–99% test-split accuracy reported in Section IV-A; we cannot yet rule out that the dev subset is harder, or a genuine reconstruction gap, and flag this as an open validation question (Section VI) rather than resolving it here. *DetectCodeGPT* [4] is a zero-shot statistic from the DetectGPT [14]/NPR [15] family, computed from CodeLlama-7B [16] and CodeT5p-770M [17], with $K=50$ perturbations per a pre-registered convergence study. Its threshold, calibrated on an independent 40-sample set, reaches only 60.0% accuracy, so raw score movement is primary evidence and threshold-derived decisions secondary. *GPTSniffer* [3] and *CodeGPTSensor/+* [5] could not be integrated, as neither released a checkpoint; we did not train replacements, to avoid confounds distinct from evaluating a published detector.

RQs: This evaluation is structured around three research questions.

- **RQ1:** How stable are detector scores and decisions under formatting, lexical renaming, and control-flow rewriting?
- **RQ2:** Do detectors exhibit similar or detector-specific stability profiles under identical transformations?

TABLE I
DROIDDETECT-BASE FORMATTING IER BY TRUE ORIGIN CLASS
(HUMAN-VS-AI COLLAPSE, POOLED ACROSS LANGUAGES).

True class	n (baseline-correct)	IER	95% CI
HUMAN_GENERATED	50	58.0% (29/50)	[44.2,70.6]
MACHINE_GENERATED	58	0.0% (0/58)	[0.0,6.2]
MACHINE_REFINED	47	4.3% (2/47)	[1.2,14.3]

- **RQ3:** Does canonical normalization attenuate transformation-associated instability, and does this differ across detectors?

IV. RESULTS

A. RQ1/RQ2: detector-specific, transformation-specific profiles

Table II reports IER with baseline-correct denominators. DroidDetect-Base’s formatting IER exceeds its lexical-rename and control-flow IER in both languages, collapsing to at or near 0% in Python; Wilson [20] 95% CIs for formatting do not overlap the other families in Python, while Java’s smaller denominators give wider, overlapping intervals. Table I breaks this down by true origin class: DroidDetect-Base’s formatting IER is 58.0% on genuinely human-written code but 0.0–4.3% on actually AI-authored code, consistently in both languages (Python 66.7%/0.0%/6.5%; Java 41.2%/0.0%/0.0%), not a no-op artifact (only 5–6% of AI-class transforms are byte-identical). A reverse check – messifying AI-class code (133 manifest rows, 96 baseline-correct after excluding 10 invalid messifications) – induces the reverse flip in only 5/96 cases (5.2%, 95% CI [2.2,11.6]). This is not intensity-matched: messify’s normalized edit ($d_{\text{text}}/\text{len}=0.064$) is smaller than formatting’s (≈ 0.14), so the 58.0%/5.2% gap confirms directionality, not a matched effect size (Section VI). Raw edit size does not otherwise separate flipped from unflipped human samples once length-normalized. $|\Delta|$ correlates weakly with d_{text} ($\rho \leq 0.30$) but not d_{AST} . Of the 29 flipped human samples, four-class output lands mostly on MACHINE_REFINED (17/29); Droid defines that class as human code modified by an LLM, not a deterministic formatter, so this is a genuine mislabel, not a defensible edge case. LLMsniffer’s low IER is not robustness evidence (67.0%/44.4% baseline accuracy, at/below majority); DetectCodeGPT’s elevated IER sits within its 30.0% noise floor (Section VI).

Answer: DroidDetect-Base’s instability is real, class-specific, directionally asymmetric (58.0% vs. 5.2%, though not intensity-matched), and overwhelmingly false-accusing under Droid’s own label definitions; the other two detectors’ apparent profiles are confounded by degenerate baseline accuracy and re-scoring noise, respectively.

B. RQ3: canonicalization intervention

The canonicalization diagnostic was applied to all 158 formatting pairs (94 Python, 64 Java). Canonicalizing both sides collapsed **108/158 (68.4%)** to byte-identical input ($\Delta=0$ by construction); the remaining **50/158** retained a residual

TABLE II
DROIDDETECT-BASE INDUCED ERROR RATE BY LANGUAGE AND FAMILY
(n = BASELINE-CORRECT DENOMINATOR, HUMAN-VS-AI COLLAPSE, SECTION VI). LLMSNIFFER (IER $\leq 8.3\%$ THROUGHOUT, BUT AT/BELOW MAJORITY-CLASS BASELINE) AND DETECTCODEGPT (IER 16.7–50.0%, WITHIN ITS 30.0% NOISE FLOOR) ARE SUMMARIZED IN TEXT; FULL PER-FAMILY TABLES ARE IN THE ARTIFACT.

Detector	Lang.	Family	IER (n)	95% CI
DroidDetect-Base	Py	formatting	26.1% (24/92)	[18.2,35.9]
DroidDetect-Base	Java	formatting	11.1% (7/63)	[5.5,21.2]
DroidDetect-Base	Py	lexical rename	0.0% (0/80)	[0.0,4.6]
DroidDetect-Base	Java	lexical rename	1.2% (1/82)	[0.2,6.6]
DroidDetect-Base	Py	control flow	0.0% (0/26)	[0.0,12.9]
DroidDetect-Base	Java	control flow	6.2% (1/16)	[1.1,28.3]

TABLE III
MEAN $|\Delta|$ BEFORE/AFTER CANONICALIZATION, BY LANGUAGE (NOT COMPARABLE ACROSS DETECTORS).

Detector	Lang.	Full set		Residual-only	
		before	after (red.)	before	after (red.)
LLMSniffer	Py (94/32)	0.027	0.003 (8.6 $\times$)	0.032	0.009 (3.5 $\times$)
LLMSniffer	Java (64/18)	0.025	0.002 (16.7 $\times$)	0.020	0.005 (3.7 $\times$)
DroidDetect-Base	Py (94/32)	0.315	0.009 (34.9$\times$)	0.394	0.027 (14.8$\times$)
DroidDetect-Base	Java (64/18)	0.161	0.003 (63.3$\times$)	0.139	0.009 (15.4$\times$)
DetectCodeGPT	Py (94/32)	0.687	0.280 (2.5 $\times$)	0.843	0.824 (1.0 $\times$)
DetectCodeGPT	Java (64/18)	0.645	0.244 (2.6 $\times$)	0.566	0.866 (0.7$\times$, increase)

difference and were rescored. Table III reports both, isolating the diagnostic’s contribution from the trivially-resolved cases.

On the residual-only subset – excluding pairs canonicalization trivially equalizes – DroidDetect-Base’s mean score movement still falls 14.8–15.4 $\times$, but this is complicated by an out-of-distribution effect: mean probability on the *true* class drops to 0.223 (66% < 0.05) against 98–99% raw accuracy (Section VI), so canonicalized input is often confidently misclassified. DetectCodeGPT shows no comparable reduction (Python 1.0 $\times$, Java 0.7 $\times$ increase), and its canonicalized scores skew toward AI (68% above threshold vs. 50–54% baseline). LLMsniffer shows a smaller, real reduction with no comparable saturation. Fig. 1 reports the same residual-only pooled factors (3.6 $\times$ /14.9 $\times$ /0.9 $\times$), avoiding the full-set numbers that 68% $\Delta=0$ -by-construction pairs would inflate.

Answer: This canonicalizer cannot cleanly answer RQ3 for DroidDetect-Base: canonicalized input is itself frequently misclassified, so the large residual-movement reduction is confounded with saturation rather than clean evidence of resolved instability; a layout-preserving normalizer is needed before this question can be settled. DetectCodeGPT’s movement is unreduced, and on Java, larger after canonicalization.

C. Corpus feasibility for deeper transformations

Local structural extraction and semantics-preserving rewriting were additionally attempted at two intensities; of 796 attempted combinations, only 37 (4.65%) were applicable (35 validated, only 4 with true differential-execution verification), reflecting Droid’s fragment-heavy composition. The resulting per-detector cells ($n=1$ –12) are illustrative only and support no confirmatory claim.

Canonicalization intervention -- residual pairs only, within-detector before/after
(each panel its own y-axis; scales are NOT comparable across detectors)

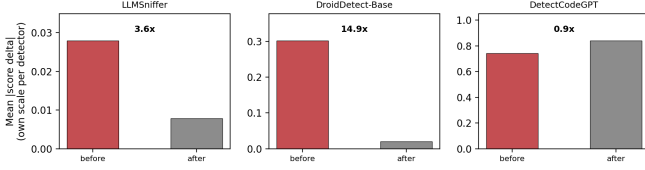


Fig. 1. Canonicalization reduction factor, residual pairs only, per detector (not cross-comparable).

V. DISCUSSION

Static accuracy does not characterize post-transformation reliability, but neither does a low IER by itself characterize stability: LLMsniiffer’s near-majority-class accuracy makes its low IER a likely degenerate-predictor artifact, and DetectCodeGPT’s IER is not yet separated from re-scoring noise. Only DroidDetect-Base’s instability survives both checks – a real detector (98–99% baseline accuracy) whose formatting sensitivity is reproducible, class-specific, and directionally asymmetric under a symmetric intervention (58.0% vs. 5.2%; Section IV-A). This asymmetry, more than the raw and imperfectly length-controlled edit-distance comparison (Section VI), is what best supports tidy formatting as a learned AI-authorship proxy, though CAST does not identify the mechanism directly. A verdict unfit to survive a random seed or a formatter run is unfit for single-verdict use. Canonicalization remains a diagnostic, not a validated mitigation: its own out-of-distribution effect means a practical fix needs a layout-preserving normalizer instead.

VI. THREATS TO VALIDITY

- **External:** Two languages, one corpus, three detectors; two others lacked a released checkpoint. Transformation operators represent common tooling edits, not sampled developer histories.
- **Construct:** LLMsniiffer’s baseline accuracy (67.0%/44.4%, at/below majority) means its low IER is not robustness evidence. Length-normalized $d_{\text{text}}/\text{len}$ is comparable (0.146 vs. 0.136) between flipped/unflipped human samples, so program length is an unexcluded confound; the messify reverse check (Section IV-A) is not intensity-matched either. DetectCodeGPT’s weak calibration limits threshold-derived results to secondary status, and its $\text{IER}/|\Delta|$ include a comparable noise floor. d_{text} , d_{token} , d_{AST} correlate only weakly with $|\Delta|$.
- **Internal:** DroidDetect-Base’s architecture was reconstructed absent public modeling code, not yet compared against the original authors’ numbers. Its 87.5% dev-split CI [69.0,95.7] does not fully contain its 98–99% test-split accuracy – an unresolved gap the entire headline finding rests on. IER uses the human-versus-AI collapse of its four-class output (16/365 intra-AI-churn flips excluded). The canonicalizer normalizes several properties

TABLE IV
NOVELTY POSITIONING AGAINST THE CLOSEST PRIOR WORK.

Work	Post-gen. edit	Routine, non-adv.	AI-code detector	Cross-det.
Detector papers [3], [2], [4], [5]	partial	partial	✓	–
CodeGPTSensor+ [6], Droid [1]	✓	– (adversarial)	✓	partial
Pordanesh et al. [11]	–	–	✓	✓
RoPGen [12]	✓	✓	– (human authorship)	–
Code-model robustness [9], [10], [7], [8]	✓	✓	–	partial
CAST (this work)	✓	✓	✓	✓

at once; canonicalized-input accuracy shows a real out-of-distribution effect not fully controlled for.

- **Corpus:** Droid’s fragment-heavy composition limits deeper transformations to 4.65% applicability and differential-execution verification to 4/35 successful instances; those cells are illustrative, not confirmatory.
- **Conclusion:** Some cells have small denominators, as low as $n=4$; Wilson 95% CIs are reported throughout. Cohen’s d_z [21] (in the artifact) is supplementary, directional-consistency evidence only.

VII. RELATED WORK

Measuring prediction change under fixed, semantics-preserving transformations is established for code models generally: Rabin et al. [9] show neural program models fail to generalize under such edits, and ReCode [10] benchmarks code-generation robustness under >30 similar operators. CAST applies the same style of edit to a different task and failure mode (attribution false-accusation, as a metamorphic relation, not degraded generation), which Table IV positions against detector papers [3], [2], [4], [5] (accuracy only, no stratification), CodeGPTSensor+ [6]/Droid [1] (adversarial, not routine), Pordanesh et al. [11] (varies generation, not the code), and RoPGen [12] (trains a *human*-authorship attributor; CAST evaluates *already-trained* detectors). Whitespace-based detection work [13] motivates why formatting sensitivity can be presentation-driven.

VIII. FUTURE PLANS

This evaluation will be extended in three directions: (1) a layout-preserving normalizer, length-controlled resampling, and an intensity-matched messify operator; (2) chained transformations on complete programs mined from commit histories; (3) comparing DroidDetect-Base against the original authors’ numbers and shared-seed DetectCodeGPT.

IX. CONCLUSION

This paper presented CAST for testing attribution stability. Its clearest finding: a reconstructed detector (real 98–99% test accuracy, though dev-split validation has an unresolved gap; Section VI) flips correct verdicts on human code to “AI” 58.0% of the time after reformatting, versus 0.0–4.3% on AI code and 5.2% under a reverse, non-intensity-matched check – consistent with tidy formatting as a learned AI-authorship proxy, and a genuine mislabel under Droid’s own MACHINE_REFINED definition. The other two detectors are confounded by baseline accuracy and re-scoring noise.

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